

Artificial intelligence-aided generative design of non-imaging secondary reflector for linear Fresnel concentrating collector

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ABSTRACT

In this article, an AI-aided generative design workflow is presented for the geometrical optimisation of secondary reflectors of linear Fresnel concentrating collectors. The geometry of the secondary reflector is based on principles of non-imaging optics, for which the exact analytical solution is derived for a specific linear Fresnel system acting as a validation test case. The artificial intelligence-aided generative design workflow is then applied on the same system and the optimal geometry generated is compared against the analytical solution. It is shown that the secondary reflector geometry obtained from the artificial intelligence-aided generative design workflow approaches the shape of the one obtained using the analytical solution and can outperform the latter with respect to the magnitude of radiative power intercepted by the receiver and radiative flux distribution at the receiver aperture.

1. Introduction

The secondary reflector in linear Fresnel concentrating collectors plays a critical role in the overall performance of the system, maximising optical efficiency and energy output [1]. The geometrical profile of the secondary reflector is designed in conjunction with the primary reflector array in such a way as to always redirect the sunlight towards the absorber that is usually mounted along the linear focal line. A comprehensive review performed by Bellos [2] emphasises that the largest portion of studies related to the overall efficiency enhancement of linear Fresnel systems are focused on the optimisation of the secondary reflector geometry. An optimally designed secondary reflector geometry should allow for optimum values of key performance metrics, such as the magnitude of radiative flux and flux distribution at the receiver aperture [3,4].

The design of the secondary reflector is a multidisciplinary process that integrates principles of optics, mechanical engineering, and manufacturing, with the goal of maximising optical efficiency and energy output while ensuring practicality, longevity of materials under thermo-mechanical stresses and cost-effectiveness. This makes the optimal design of secondary reflector a challenging task, with a non-existing established procedure to verify that the complex design space is fully explored to converge to the optimal geometry. Thus, traditional design methods focus on developing the secondary reflector, using the relevant key performance metrics discussed above as benchmarks.

Through an iterative refinement process, the geometry of the secondary reflector is adjusted on the basis of performance evaluations until it converges to a design solution that meets the desired performance criteria. Of course, in addition to performance considerations, compatibility with manufacturing processes must also be taken into account during the design process.

Over the last couple of decades, the convergence of artificial intelligence (AI) and generative design methodologies has considerably advanced engineering design processes in various domains [5]. By introducing AI algorithms and machine learning techniques, large design spaces can be explored, and optimal solutions that would otherwise be impossible to consider can be identified. Inevitably, given the potential of such methodologies, unprecedented opportunities have arisen for the design and optimisation of secondary reflectors in linear Fresnel systems offering limitless possibilities.

The main advantage of Linear Fresnel Reflectors (LFR) is in its stationary receiver enclosed by a secondary reflector (as illustrated in Fig. 1), which, when co-designed with the primary optics, allows for a better distribution of the solar flux around the absorber tube, avoiding thermal stresses due to localised hot spots and thus structural failure as described in Chaitanya Prasad et al. [6]. This is paramount to extend the lifespan of the tubes used to collect the solar light reflected by the mirror set. When working with a heat transfer fluid (HTF) without phase change, the receiver encloses a single linear tube, running across the longitudinal axis of the collector along its focal line. The usual

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Nomenclature

Latin letters

C	Absorber pipe centre
H	Height (m)
I	Involute point
i, n	Integer indices
L	Length (m)
M	Primary reflector mirror edge point
N	Integer number
q	Radiative power (W)
q''	Radiative flux (W m^{-2})
r	Absorber pipe radius (m)
$S_{q''}$	Radiative flux non-uniformity (W m^{-2})
x	Decision variable in the NSGA-II
X, Y, Z	Cartesian coordinates of dual-pipe absorber (m)
x, y, z	Cartesian coordinates of single pipe absorber (m)

Greek letters

θ	Angle (rad)
ψ	First control point angle in the cubic b-spline interpolation ($^\circ$)

Subscript

1	Right side of NIO analytical design
2	Left side of NIO analytical design
a	Acceptance
abs	Absorber
an	Analytical
ap	Aperture
int	Intercepted
f	Discrete surface element
ray	Ray in Monte-Carlo ray tracing
s	Sun

Abbreviations

AD	Analytical design
CPC	Compound parabolic concentrator
CSR	Circumsolar ratio (rad)
DNI	Direct normal irradiation (W m^{-2})
HTF	Heat transfer fluid
LFR	Linear Fresnel reflector
NIO	Non-imaging optics

practice for designing the secondary reflector of such a system is to employ designs of compound parabolic collectors (CPC). Scientific work has been carried out in numerous initiatives to optimise these shapes, focusing on increasing the light collection and concentration efficiency, and harmonising the flux distribution around the absorber tube. For example, for a given absorber tube diameter, analytical solutions were developed as discussed in McIntire [7]. Cheng et al. [8] focused on a coupled optimisation of the primary reflector (mirror width, distance between mirrors and mirror shape), CPC (position, acceptance angle and opening angle) and absorber (size and position) for the single objective of maximising the optical efficiency of the overall system. The same work was later expanded in Men et al. [9] as a bi-objective problem of maximising the optical efficiency and minimising the radiative flux non-uniformity of the system, obtaining optical efficiencies around 60%. Furthermore other optimisation procedures focusing on a CPC shape for the secondary reflector, led to efficiencies up to 84% [10,11].

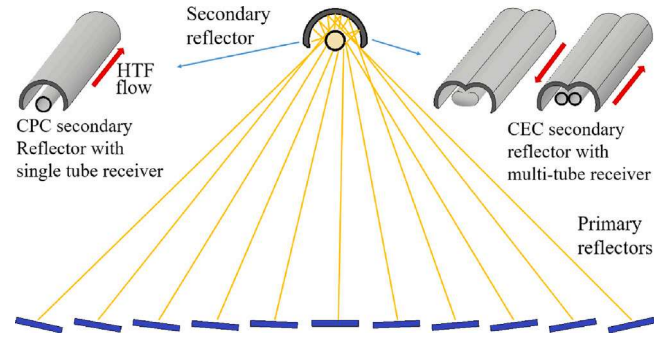


Fig. 1. Typical layout of an LFR with its CPC secondary reflector. Notice that for a multi-tube absorber with a cavity-type secondary reflector, the absorber has a U-bent at one end in order for the HTF to return to the same side as the inlet, reducing thus the HTF piping that is exposed outdoor.



Fig. 2. An elevated view of CyI's linear Fresnel collector.

With regard to the absorber, another approach that is proposed for linear Fresnel collectors is the so-called multitube linear absorber, in which several tubes are used in parallel under a single secondary reflector geometry. For such configurations, a trapezoidal-type secondary reflector geometry that significantly improved the thermal efficiency was presented by Abbas et al. [12]. Similarly, the present study focuses on a multi-tube absorber structure, in which two tangent tubes are running side by side along the longitudinal axis of the Fresnel collector. At one end of the collector, the two tubes are connected to each other by a torus as shown in Fig. 1, so that the HTF inside the tubes is counter-flowing, thus entering and exiting the Fresnel collector at the same side. For this kind of absorber configuration, an optimised cavity-type secondary reflector geometry is necessary to improve the flux distribution around the absorber tubes.

This article presents a novel AI-aided generative design workflow for optimising the geometry of the secondary reflector for such a multi-tube absorber geometry. By integrating AI algorithms coupled to generative design methods, the workflow overcomes the limitations of traditional design approaches, generating secondary reflector geometries that can achieve superior performance, enhancing the overall optical efficiency of the system. As a test case, the workflow is applied to optimise the secondary reflector of the operating LFR at the Cyprus Institute (CyI) for a given absorber geometry. The optimised secondary reflector generated is benchmarked against the analytical solution obtained for the same exact system and absorber geometry. It is shown that the secondary reflector geometry obtained from the AI-aided generative design workflow approaches the shape of the one resulted by the analytical solution, and outperforms the latter with respect to the set objective functions.

2. Problem statement

2.1. System description

Fig. 2 shows the linear Fresnel concentrating collector at the Cyprus Institute. The primary reflector field is composed of 288 mirrors, driven by 72 DC motors that track the sun, with a global primary reflector area of 184.32 m^2 . In the current configuration, the HTF flows into

a single 32 m long absorber pipe, entering from the south side and exiting at the north side. Mineral oil is used as HTF [13]. For this configuration, a CPC-type secondary reflector is used as shown in Fig. 1. The thermo-optical performance of the current configuration was evaluated by Montenon et al. [14] and an upgrade of the current facility was studied by Montenon et al. [15] in an effort to increase the efficiency of the collector and improve the distribution of the solar flux intercepted by the absorber surface. For the latter, an optimised non-imaging secondary reflector is designed analytically for a multi-tube absorber configuration, as schematically shown in Fig. 1.

2.2. Rationale for non-imaging optics

While imaging optics systems are focused on replicating images with sharpness and fidelity, non-imaging optics systems aim to control and tune radiation distribution for various functional and specific purposes such as illumination, concentration and diffusion without necessarily reproducing detailed images [16]. In LFR systems, the secondary reflector can be geometrically co-designed with the primary reflector. Thus, the flux can be better harmonised around the absorber, which is important in order to avoid thermal stresses in the absorber wall. The primary optics distributes the reflected radiation on the lower part of the absorber and towards the secondary optics. The secondary optics harmonises the solar reflected flux on the top of the absorber. Applying non-imaging optics to reach the optimal design of the secondary reflector offers several advantages through the achievement of:

- uniform illumination by distributing radiation evenly across the outer surface of the absorber, minimising thermal gradients and leading to improved efficiency;
- wider acceptance angle, allowing maximisation of the captured sunlight over greater a range of incident angles, thus maintaining higher efficiency over the annual solar incidence angles;
- higher tolerance to misalignments occurring in assembly of such systems, i.e. deviations in the position or orientation of the primary reflector or secondary reflector, thus improving the robustness of the system and reducing the need for precise alignment during installation and operation;
- reduced optical losses due to reduced spillage, reflections, scattering, and absorption of sunlight, improving the overall efficiency by maximising the amount of sunlight delivered to the receiver;
- more compact and lightweight design of the secondary reflector compared to a cavity-type solution, beneficial for easing the installation and transportation, and increasing the cost-effectiveness of the overall system.

3. Analytical design

Given the primary mirror configuration and the same height of the absorber of CyI's actual LFR, an upgraded version of the secondary reflector and absorber subsystem is derived analytically. A dual-pipe receiver with a U-bent shape at the south end is considered, aiming at reducing losses of ambient piping exposure (see Fig. 1). Non-imaging optics is applied to the new secondary optics to smoothly distribute the solar flux around the absorber, to improve concentration, and overall to achieve a higher concentration–acceptance product (CAP) [16]. For the current study, studying the physics within the U-bent section of the absorber is beyond the scope of this paper.

The secondary reflector is generated via the use of two symmetric involutes and two symmetric macro-focal ellipse (MFE) bundles as shown in Fig. 3(b).

The solar angle radius stemming from the sun, is θ_s , which is close to 4.7 mrad. Its summit point M is the furthestmost point of the primary optics set from the receiver, being at the edge of the furthest mirror (on the left side, in the present case). The acceptance angle is θ_a with same summit, but pointing to the receiver. The angle θ_a is wider than θ_s ($\theta_a =$

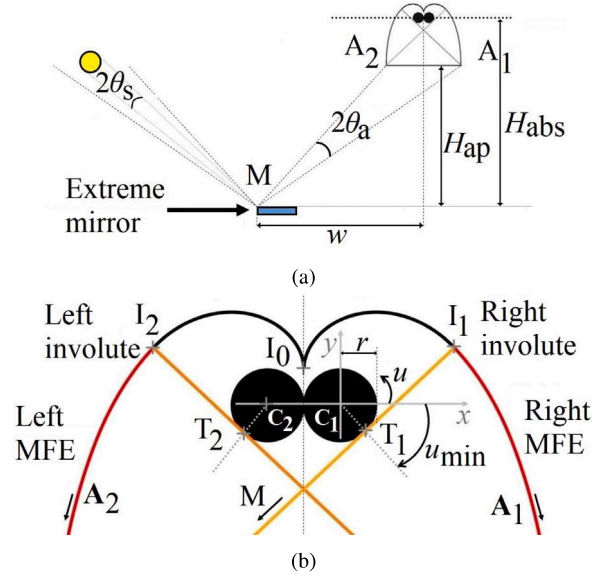


Fig. 3. NIO analytical secondary reflector geometry (a) extreme edge of the furthestmost mirror and receiver. (b) Close-up view of the receiver and its composing elements.

$2\theta_s$, more precisely), to accommodate tracking errors and to ensure that the inner mirrors will reflect within the aperture width, since non-imaging optics is applied onto a pre-existing primary optics [15]. The two absorber pipes are represented in black circles with centres C_1 and C_2 for the right side left sides, respectively. The centers of the absorbers are located at the same height H as CyI's Fresnel collector. On the schematics, index 1 refers to the right half, and 2, the left one. The top edge of θ_a is tangential to the right absorber at the point T_1 that defines the following system:

$$\begin{cases} \overline{MT_1} \cdot \overline{C_1T_1} = 0 \\ \overline{C_1T_1}^2 = r^2 \end{cases} \quad (1)$$

The system has two solutions. Thus, the following conditions should be satisfied for $T_1(x, y)$: $y < H_a$.

Once T_1 is defined, the line (MT_1) sets the boundaries of the right and left involutes at the points I_1 and A_2 , respectively. The two involutes are generated from the convex hull of the two tubes. The right involute starts the point I_0 with coordinates $(-r, r)$ with C_1 as the origin and ends on I_1 . Involute are defined by the following system of equations:

$$\begin{cases} x = r[\cos(u) + (u - \frac{\pi}{2})\sin(u) + \cos(u + \frac{\pi}{2})] \\ y = r[\sin(u) - (u - \frac{\pi}{2})\cos(u) + \sin(u + \frac{\pi}{2})] \\ z = v \end{cases} \quad (2)$$

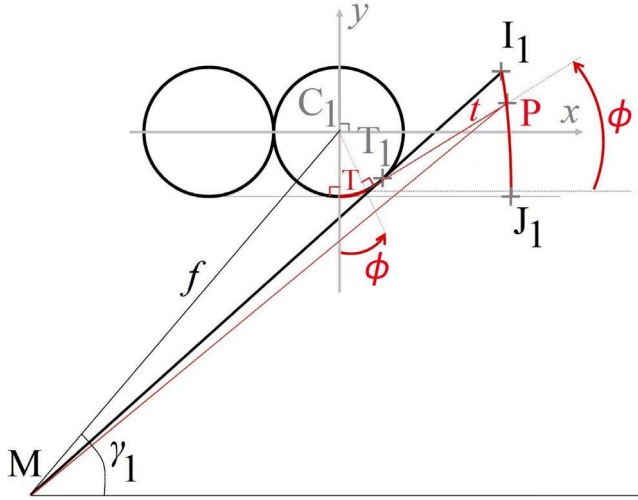
where (x, y, z) is the triplet of coordinates of points of the right involute, $u \in [u_{min}, \frac{\pi}{2}]$ and $v \in [0, L_{abs}]$.

The right involute stops its generation at the point T_1 , corresponding to the point I_1 . The angle u at this point is $u_{min} = \arctan(\frac{x}{y})$, where (x, y) corresponds to the coordinates of T_1 obtained in system (1).

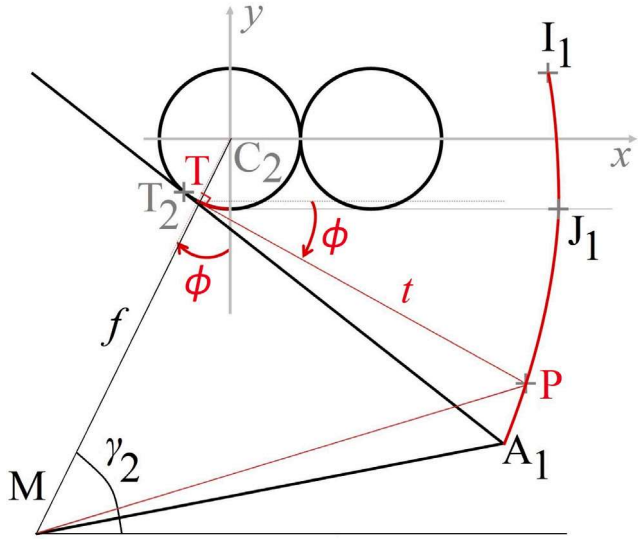
The lower part of the new secondary optics is formed on both sides by two unwinding macro-focal ellipses (MFE) for each side, as defined by Chaves [16]. M is the extreme point from which the MFE is generated. The top MFE is generated starting from the point I_1 . The focus follows the outer wall of the absorber with centre C_1 . Any point P belonging to the MFE can be defined as:

$$\overline{MP} = \overline{MC_1} + \overline{C_1T_1} + \overline{T_1P} \quad (3a)$$

$$\overline{MP} = \begin{cases} f \cos(\gamma_1) + r \sin(\phi) + t \cos(\phi) \\ f \sin(\gamma_1) - r \cos(\phi) + t \sin(\phi) \end{cases} \quad (3b)$$



(a)



(b)

Fig. 4. Details of the MFE geometry.

where $f = \|\overline{MC_1}\|$, and the angles ϕ and γ_1 are defined in Fig. 4. According to the definition of MFE:

$$t + s + r\phi = K, \quad s = \|\overline{MP}\| \quad (4)$$

where K is calculated for I_1 :

$$K = \|\overline{MI_1}\| + \|\overline{I_1T_1}\| + r \left(\frac{\pi}{2} + u_{\min} \right) \quad (5)$$

Thus,

$$\overline{MP}^2 = s^2 = f^2 + r^2 + 2fr \sin(\phi - \gamma_2) + 2ft \cos(\phi - \gamma_2) \quad (6)$$

This brings to the definition of t as a function of the angle ϕ :

$$t(\phi) = \frac{(K - r\phi)^2 + 2fr \sin(\phi - \gamma_1) - f^2 - r^2}{2(K - r\phi + f \cos(\phi - \gamma_1))}, \quad \phi \in \left[0, \frac{\pi}{2} + u_{\min} \right] \quad (7)$$

The top MFE (see Fig. 4(b)) is defined by the following system of equations:

$$\begin{cases} x = r \sin(u) + t(u) \cos(u) \\ y = -r \cos(u) + t(u) \sin(u) \\ z = v \end{cases} \quad (8)$$

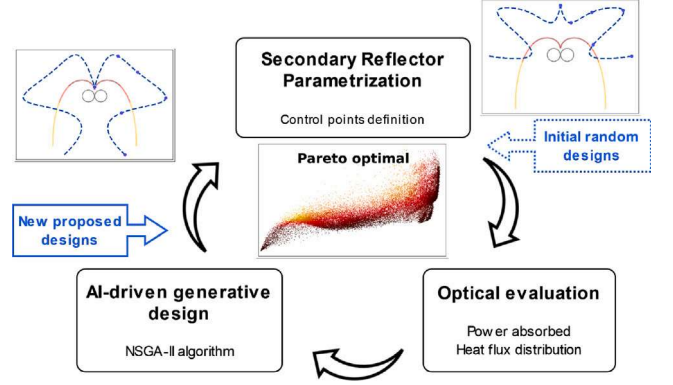


Fig. 5. The AI-aided generative design workflow adopted for the geometrical optimisation of the secondary reflector.

where (x, y, z) is the triplet of coordinates of points of the right top MFE, $u \in [0, \frac{\pi}{2} - u_{\min}]$ and $v \in [0, L_{abs}]$. The top MFE extends to J_1 where $\phi = 0$. From this point onwards, a new MFE is generated with the absorber wall on the left and with center C_2 as the focus. As the MFE passes by J_1 :

$$K = \|\overline{MJ_1}\| + \overline{C_1J_1} \cdot \bar{X} + 2r \quad (9a)$$

$$K = s + t - r\phi \quad (9b)$$

Any point P belonging to the lower MFE (see Fig. 4(a)) can be defined as:

$$\begin{aligned} \overline{MP} &= \overline{MC_2} + \overline{C_2T_2} + \overline{T_2P} \\ &= \begin{cases} f \cos(\gamma_2) - r \sin(\phi) + t \cos(\phi) \\ f \sin(\gamma_2) - r \cos(\phi) - t \sin(\phi) \end{cases} \end{aligned} \quad (10)$$

Thus,

$$\overline{MP}^2 = s^2 \quad (11a)$$

$$\overline{MP}^2 = f^2 + r^2 - 2fr \sin(\phi + \gamma_1) + 2ft \cos(\phi + \gamma_1) \quad (11b)$$

$$t(\phi) = \frac{(K + r\phi)^2 + 2fr \sin(\phi + \gamma_2) - f^2 - r^2}{2(K + r\phi + f \cos(\phi + \gamma_2))}, \quad \phi \in \left[0, \frac{\pi}{2} + u_{\min} \right] \quad (12)$$

The lower MFE is defined with the following system of equations, relatively to the center C_2 :

$$\begin{cases} x = -r \sin(u) + t(u) \cos(u) \\ y = -r \cos(u) - t(u) \sin(u) \\ z = v \end{cases} \quad (13)$$

where (x, y, z) is the triplet of coordinates of points of the right top MFE, $u \in [0, \frac{\pi}{2} - u_{\min}]$ and $v \in [0, L_{abs}]$. The lowest and top MFEs are reproduced symmetrically on the left side to complete the secondary reflector geometry. The lowest MFE ends at the points A_1 and A_2 respectively.

4. AI-aided generative design

The AI-aided generative design workflow developed for the optimisation of the geometry of the secondary reflector is shown in Fig. 5. The process starts with the definition of the Cyl linear Fresnel configuration and includes the definition of all parameters that constitute the basic system, that is, the primary reflector along with its properties (mirror reflectivity and shape, tracking characteristics, etc.), atmospheric conditions, and the position of the sun along with the corresponding direct normal irradiation at a given time or at a set of

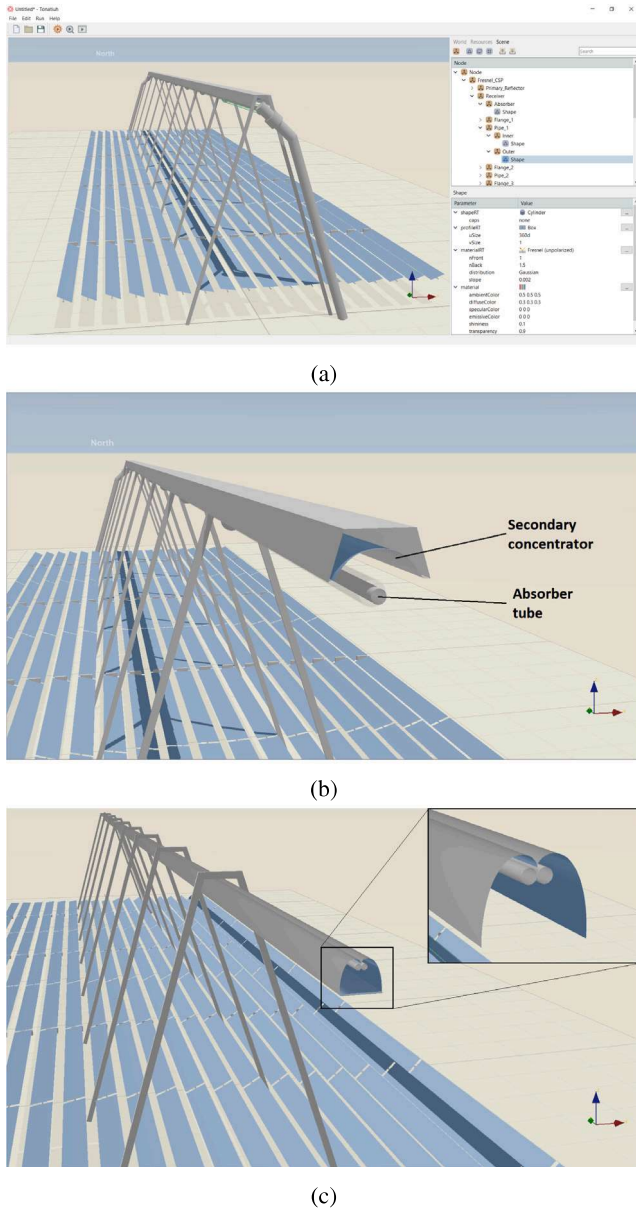


Fig. 6. Monte-Carlo ray-tracing model system rendered with Tonatiuh++, developed as an exact replica of Cyl's LFR: (a) overall view, (b) close-up view of the initial secondary reflector and single absorber tube, and c) close-up view of the non-imaging optics secondary reflector and the dual-pipe receiver.

sun positions. It includes also an initial geometry parametrisation of the secondary reflector to be optimised. Thus, a Monte-Carlo ray-tracing scene that fully replicates the basic system configuration is automatically constructed, in which the optimised secondary reflector optical performance is simulated. Tonatiuh++ ray tracer [17,18] is integrated into the workflow for this step. After optical simulation, the AI part of the workflow comes into action, where a non-dominated sorting genetic algorithm II (NSGA-II) takes over for multi-objective multi-variable evaluation of the candidate secondary reflector geometries according to the set objective functions. After evaluation, a new secondary reflector design population is automatically generated and fed back into the initial step for re-evaluation. The result is a multidimensional Pareto front with candidate secondary reflector geometries, corresponding to the set objective functions.

4.1. Monte-Carlo ray-tracing model system

Fig. 6(a) shows the Monte-Carlo ray-tracing model system rendered with Tonatiuh++ as an exact replica of the Cyl's LFR, while Fig. 6(b) shows the existing secondary reflector and absorber arrangement. Using the analytical geometry model of the upgraded version of the secondary reflector presented in Section 3, the exact geometrical model is incorporated into Tonatiuh++ and integrated with an existing model of Cyl's LFR, as shown in Fig. 6(c) [19]. To perform comparative optical performance analyses between the secondary reflector geometries obtained using the AI-aided generative design detailed next and the analytical design detailed in the previous section, Monte-Carlo ray-tracing simulations have been carried out for both geometries.

In LFR systems, the theoretical distribution of the flux on the receiver's surface has the same linear profile all along its length due to the longitudinal cosine effect, except at the tube inlet and outlet where the end effects occur [20]. Thus, for the purpose of the present study and to maintain consistency with the analytical solution, the following assumptions are considered to reduce the computational cost of the AI-aided generative design approach:

- In the ray-tracing scene as presented in Fig. 6(c), only the critical surfaces are kept. The secondary reflector structure, tracking motor boxes, beams, and supports of the primary reflector, as well as the U-bent and external pipes of the HTF circuit, are omitted to reduce the amount of surfaces involved in ray tracing. The inevitable shadows projected by the structure on the primary reflector are considered to be negligible due to the small size relative to the size of the primary reflector.
- In the absence of a secondary-axis tracking system, the cosine losses in LFRs are inevitable, because they are intrinsic to such systems. Considering these characteristics, only a reduced version in length of the LFR can be used to allow a significant reduction of the rays used in simulations. To maintain the collector fully irradiated while minimising the cosine losses, the sun's elevation is kept at 90° , so that the 3D simulation can be effectively reduced to a pseudo-2D simulation.

Implementing the above points results in the modified ray-tracing scenes as shown in Fig. 7. In the details of the simulations, the Buie sunshape was selected with a circumsolar ratio (CSR) of 2%, while the slope error distribution of the different reflective surfaces was set to normal with a standard deviation of 2 mrad. The reflectivities of the primary and secondary reflectors were set to 92.4% and 88.3%, respectively, which correspond to values of the currently operational collector. For both cases, the external surfaces of the absorber tubes have been treated as perfect absorbers. For the simulations, a single-point aiming strategy was selected. This corresponds to the center of the inlet aperture glass, maintaining thus the same configuration as used in the analytical solution, i.e. 13.5 cm below the horizontal centerline of the dual-pipe absorber. The optical length between the primary reflector and other elements of the LFR collector is sufficiently small to justify neglecting the effects of atmospheric attenuation. Based on a ray convergence study, a total of 10^7 rays were necessary to irradiate the reduced LFR collector. For all simulations, direct normal irradiation (DNI) was set equal to 1000 W m^{-2} . Fig. 8 shows example rendering of ray tracing in the reduced collector.

4.2. Geometry parametrisation

For the parametrisation of the geometry of the secondary reflector, polar coordinates are employed, in which sampled control points are used to interpolate and generate the profile of the secondary reflector. Due to symmetry and the risk of having a singularity in the secondary reflector profile, the control points only define half of the profile, which is later mirrored. The arrangement of the control points is defined by

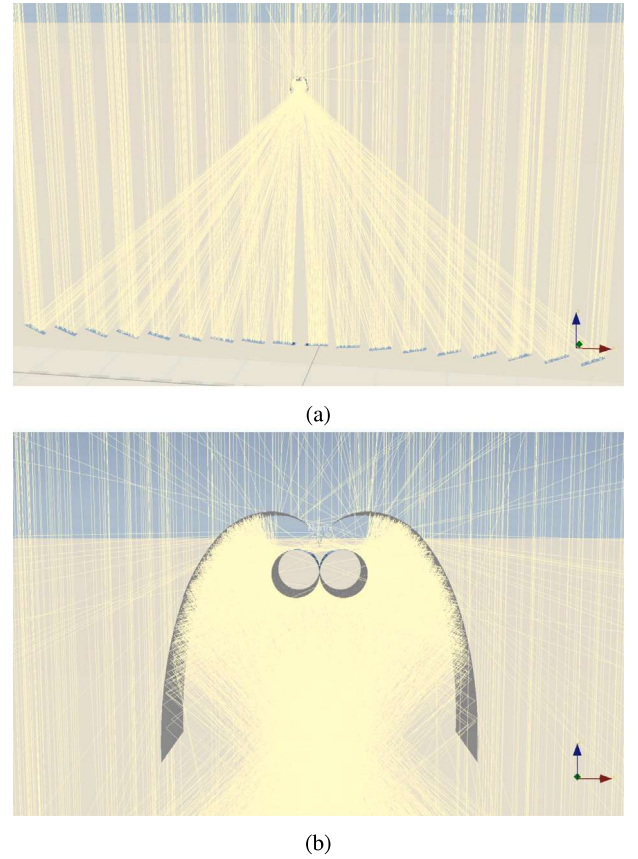
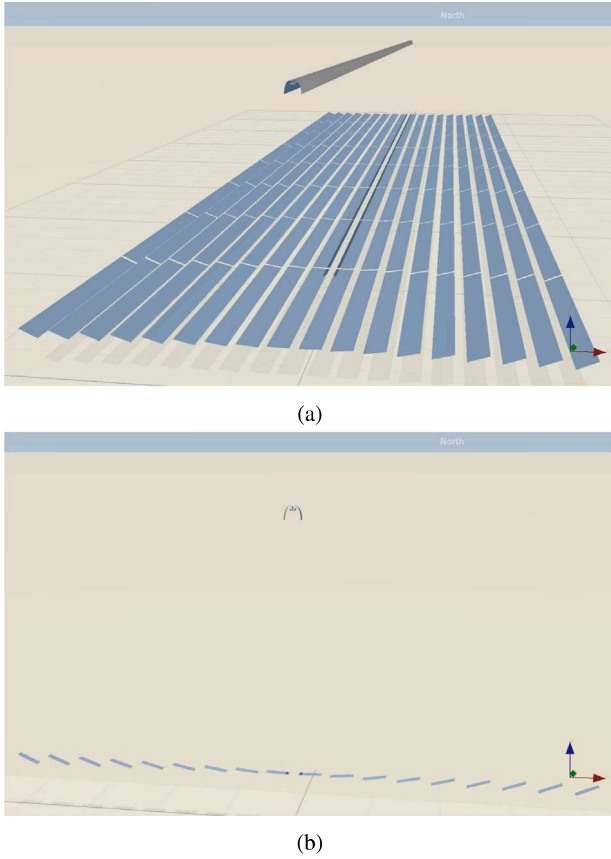


Fig. 7. Reduced ray tracing scene: (a) keeping only the critical surfaces, and (b) reduction of the length to 0.1 m due to the 2D nature of the problem, sun position at 90° elevation and 180° azimuth.

Fig. 8. Example rendering of ray tracing for sun position at 90° elevation and 180° azimuth: (a) the reduced length collector, (b) Close-up view of the dual-pipe receiver and the non-imaging analytical design secondary reflector.

the angle with respect to the reference system of the first control point ψ , and the corresponding distances of all control points to the origin of the reference system d_{1-n} . The angular position of the control point n is always considered to be $\frac{\pi}{2}$ due to the assumed symmetry conditions, and the angular position of the remaining control points is the one that makes all control points equiangular between the first and last control points. An example can be observed in Fig. 9(a). The secondary reflector profile is generated by interpolating the control points in a cubic B-spline curve following the methods detailed in Dierckx [21] and implemented by Virtanen et al. [22], without any smoothing conditions to force the control points to coincide with the spline as shown in Fig. 9(b).

Multiple secondary reflectors can be designed using the described parametrisation. The resolution of the designed curve depends on the number of control points employed in the interpolation and the given range in which each control point displaced. Therefore, it is necessary to properly define these values to reduce combinatorics to meaningful options, thus reducing computational cost and allowing for faster convergence of the algorithm explained later in Section 4.3.

Considering the non-imaging optics secondary reflector design as derived in Section 3, the displacement range of the control points distances is defined in the range of (0,0.2] m. The displacement range for the angular variable ψ is set to $[-\frac{\pi}{2}, 0.2]$ rad to allow the geometry of the secondary reflector to become a closed curve or an open curve starting above the dual-pipe collector.

The number of control points used in the interpolation is a key factor for the definition of the secondary reflector geometry, because more points result in higher fidelity and complexity of the generated reflector geometries. This in turn allows for more combinatoric evaluations. Fig.

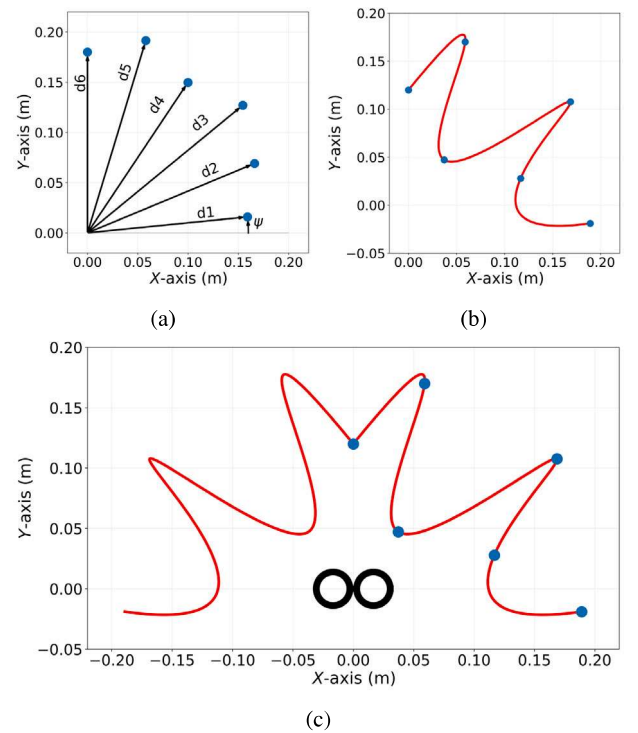


Fig. 9. Example of control point arrangement and cubic B-spline interpolation curve for $n = 6$ points: (a) control point arrangement, (b) cubic B-spline interpolation curve, and (c) mirrored cubic B-spline interpolation curve to generate the full secondary reflector profile.

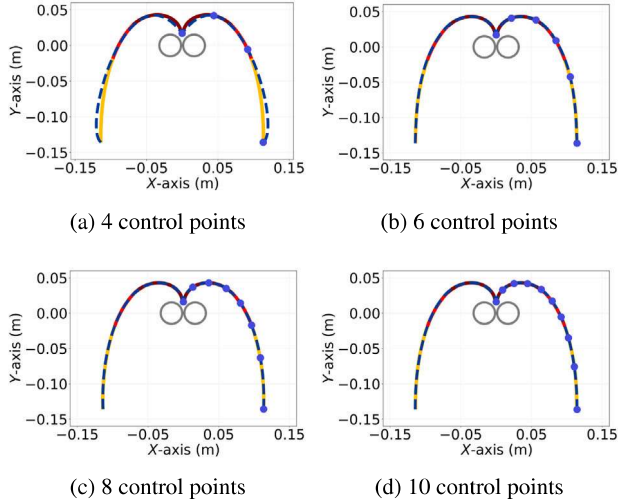


Fig. 10. Non-imaging optics replicability of the secondary reflector geometry using cubic B-spline interpolation curves for different number of control points.

10 illustrates examples of the number of control points that is necessary for the B-spline interpolation curve to efficiently approximate the geometry of the secondary reflector geometry as derived analytically. The minimum number of six required points is used in all parametrised designs employed in the present study.

As pointed out earlier in Section 4.1, the aimpoint location is fixed at the center of the inlet aperture glass (at $X=0$), and at a distance of $Y=-13.5$ below the horizontal centerline of the dual pipe absorber. Although the aimpoint could be allowed to adapt to each generated geometry, meaning that the vertical coordinate of the first control point of the reflector geometry would be colinear to the Y-coordinate of the aimpoint, the present study employs a fixed aimpoint location. This allows the optimisation workflow to focus solely on the geometry of the secondary reflector without optimising simultaneously the aimpoint location.

4.3. Non-dominated sorting genetic algorithm II

Genetic algorithms (GA) are search and optimisation algorithms inspired by the principles of genetics and aiming to replicate the biological processes of natural selection and adaptation to environmental conditions. GAs are particularly well suited to solve complex problems with large search spaces. Holland [23] established the key components and principles of a genetic algorithm, which includes fitness evaluation, parental selection, cross-over, mutation, and inversion processes. A population of individuals is generated and then evaluated based on specific objectives during the fitness evaluation process. Individuals with higher fitness are prioritised to pass their genes to the next generation in the parental selection process. Once the parents are selected, their genes are combined during the crossover to generate the offspring. In the mutation process, some genes of the offspring are randomly altered, increasing the diversity of the new population. The last process is inversion, which randomly inverts the value of some genes which are treated as a boolean variable.

A variant of GA, the non-dominated sorting genetic algorithm (NSGA), was presented by Srinivas and Deb [24] to better address multiobjective optimisation problems. NSGA incorporates the concept of non-dominated sorting to rank the best solutions in a population. An individual is considered to dominate if it outperforms the rest of the population in one objective while performing at least as well in all other objectives, so that the best solutions are the non-dominated ones. In that case, the first front is composed of all the individuals that are not dominated in the population; the second front are the non-dominated

solutions in the population when the first front individuals are extracted from the population; the algorithm continues until all the individuals are sorted in multiple fronts. Based on NSGA, two more variations NSGA-II and NSGA-III were proposed in Deb et al. [25] and Deb and Jain [26], respectively.

The development of NSGA-III was motivated by the need to perform many-objective optimisations, validated in normalised test problems with a number of objective functions ranging between 3 and 15. However, the LFRs optimisation problem presented in this article is bounded with only two objectives functions, and thus the NSGA-II is deemed to be a better approach. For the sake of clarity, an individual is a secondary reflector design, and a population is the group of designs considered by NSGA-II in every generation composed of the parents (designs with best objectives functions) of the previous generation and the offspring (new generated designs).

In an optimisation problem, the goal is to find the values of one or more variables (x_{1-n}) for which the objective functions satisfy the set criteria, within appropriate constraints. In the present study, the power intercepted by the dual-pipe receiver is maximised and the non-uniformity of the radiative flux on its surface minimised. These objectives are translated into two objective functions evaluated by the NSGA-II algorithm.

The first objective function is the power intercepted by the dual-pipe receiver, which is calculated as

$$q_{\text{int}} = q_{\text{ray}} \sum N_{\text{rays,int}} \quad (14)$$

where q_{ray} and $N_{\text{rays,int}}$ are the power of a single ray and the number of rays intercepted by the receiver, respectively.

The second objective function is the radiative flux non-uniformity, which is evaluated as the standard deviation of the radiative flux along the perimeter of the dual-pipe receiver,

$$S_{q''} = \sqrt{\frac{\sum_{i=1}^{N_f} (q_i'' - \bar{q}'')^2}{N_f}} \quad (15)$$

where q_i'' is the radiative flux on a surface element i , $\bar{q}''$ is the mean radiative flux along the perimeter, and N_f is the total number of discrete surface elements (faces) $i = 1 \dots N_f$ by which the geometry of the secondary reflector is represented in the Monte-Carlo ray-tracing simulations. To increase the efficiency of NSGA-II, additional considerations were made during its implementation. One single front can be composed of numerous individuals (i.e. secondary reflector designs), but with undesired characteristics. For example, in some cases a secondary reflector can be intersected by a low number of rays producing a uniform radiative flux distribution at low power intercepted. Such designs are candidate individuals on the first front. To avoid their propagation, an *inequality constraint* is implemented, in which only the designs offering intercepted power greater than a prescribed minimum level are allowed to become future parents. The minimum level of intercepted power that a candidate design must result is at least 90% of the power intercepted for the case with the secondary reflector geometry obtained using the analytical design. The power intercepted when using the analytical design is set based on results of auxiliary Monte-Carlo ray-tracing simulations using the geometry of the secondary reflector obtained with the analytical design for the reduced Fresnel collector as described in Section 4.1 and depicted in Fig. 8(b),

$$q_{\text{int}} \geq 0.9q_{\text{an}} \quad (16)$$

The secondary reflector design space is defined by the bounding area of the control points discussed in Section 4.2 and shown in Fig. 9(a). The angle ψ and the distance from a control point to the origin of the coordinate system are the first ($n = 1$) and remaining ($n = 2, \dots, N$) decision variables, respectively, used by the NSGA-II algorithm. The variable bounds are given by

$$-\frac{\pi}{2} \leq x_n \leq 0.200 \quad \text{for } n = 1 \quad (17a)$$

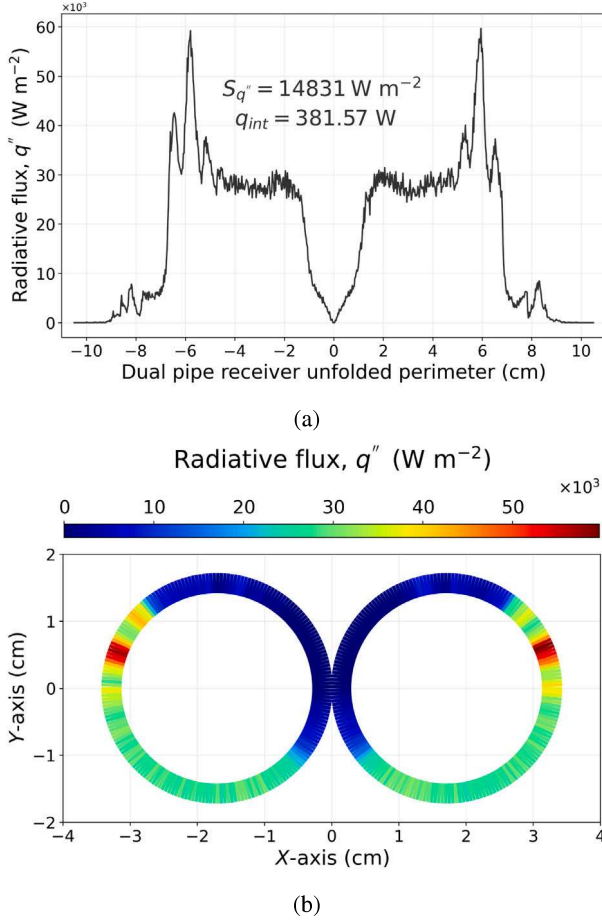


Fig. 11. Radiative flux distribution on the surface of the dual-pipe receiver using the analytically-designed secondary reflector with 88.3% reflectivity for sun position with elevation of 90° and azimuth of 180° shown: (a) along the unfolded perimeter of the dual-pipe receiver, and (b) at the actual surface of the dual-pipe receiver.

$$0.001 \leq x_n \leq 0.200 \quad \text{for } n = 2, \dots, N_x \quad (17b)$$

The implementation of NSGA-II necessitates setting up problem-specific hyperparameters [27]. In the present study, a single set-up is sufficient to achieve the objectives of this study while using the computationally costly Monte-Carlo ray-tracing simulations as the evaluation function. The hyperparameters are established to maximise the geometry diversity in the population and exploration to avoid the algorithm trap in local minima. Based on other examples in the literature [28–30], a parent population of 200 individuals and an offspring of 150 are set up, meaning a total population of 350. During the reproduction process (generation of the offspring individuals), the probabilities used in the crossover and mutation operators are set to 0.6 and 0.7, respectively.

5. Results and discussion

5.1. Performance evaluation: analytical design

The optical performance of the system employing the analytically-designed non-imaging secondary reflector is evaluated to provide the reference for evaluating the performance of the reflector designed using the AI-aided generative algorithm. The radiative flux distribution obtained for the non-imaging optics secondary reflector is depicted in Fig. 11 [15]. The power intercepted by the dual-pipe receiver for the reduced Fresnel collector configuration is $q_{\text{int}} = 381.57 \text{ W}$, while the radiative flux non-uniformity is $S_{q''} = 14831.33 \text{ W m}^{-2}$.

5.2. Performance evaluation: AI-aided generative design

The optical performance of the LFR system with the AI-aided generative-designed receiver is shown in Fig. 12. The NSGA-II evolution over the generations is illustrated in Fig. 12(a), where the set objective functions of the population can be observed for each generation. The figure captures how the designs evolved towards maximisation of power and minimisation of the radiative flux non-uniformity. Each point in the graph represents a single secondary reflector design, where the colouring of each point implies the generation in which the design is generated. The performance of the analytically-designed reflector is shown as a red star marker.

The vertical dashed line in Fig. 12 corresponds to the inequality constraint set that provoke the algorithm to discard all the low-power intercepted designs, focusing only on the designs that allow a power intercepted close to the figure of merit. The complete objective function space of all the proposed secondary reflector designs (each dot corresponds to a proposed reflector design) is shown in Fig. 12(b). The colour of each dot indicates the stage of the NSGA-II evolution at which the corresponding design was generated. For example, the first secondary reflector geometries (i.e. the first parents) correspond to the 0–199 proposed designs, which are coloured in deep purple colour. In every generation 150 new designs are proposed, thus during the second generation, the 200–349 proposed designs are evaluated. The main difference with respect to Fig. 12(a) is that over the generations, the NSGA-II maintains only the best performing designs in the population, discarding the proposed designs, if these do not outperform the parents. It is evident that although at specific regions of the plot there is an increase of generated reflector candidates, i.e. regions on the plot with higher point density, designs also exist with bad performance in the last generations, which are discarded by the NSGA-II.

Fig. 12(c) shows the Pareto optimal front as a set of efficient reflector geometry choices in yellow dots. Although the Pareto optimal front gives information about the performance of optimal secondary reflector geometries, a deeper look into the generated geometries is deemed necessary. Given the secondary reflector designs in the Pareto optimal front, it can be seen that the NSGA-II is converging into two designs (shown in Section 5.3) that are competing against each other. This is evident in Fig. 12(c), in which two Pareto sub-sections are clearly distinguished. In the first Pareto sub-section, named as the AI-AD-like, secondary reflector designs are generated with their geometry closely approaching the shape of the one obtained using the analytical design. The second Pareto sub-section, named as the AI-bullet-like, secondary reflector designs are generated with their geometry approaching a bullet shape design.

The AI-AD-like designs provide the highest values of the power intercepted in the dual-pipe receiver among all evaluated AI-aided generatively-designed secondary reflectors, but result in a more non-uniform radiative flux distribution. On the other hand, AI-bullet-like topology designs exhibit a more uniform radiative flux distribution at the expense of intercepting less power, i.e. of the order of $\sim 5\%$ less power intercepted with the analytical design.

Most of the designs located on the Pareto front outperform the analytical designs in terms of the radiative flux distribution. Furthermore, some Pareto AI-AD-like designs outperform the analytical design in terms of the power intercepted by the receiver, even if only by a small margin of $\sim 1.4\%$. This difference in the intercepted power is attributed to the fact that the analytical design has a slightly bigger aperture, which projects a small shadow on the primary reflector, while such a wide aperture is not needed to reflect the incident sunlight onto the receiver surface. The latter is explained by the principles underpinning the analytical design, in which the acceptance angle is defined as double the solar angular width to ensure that all rays reach the secondary reflector surface, even in the presence of material imperfections in the primary reflector or tracking errors. The more uniform radiative flux distribution of the AI designed receivers is a consequence

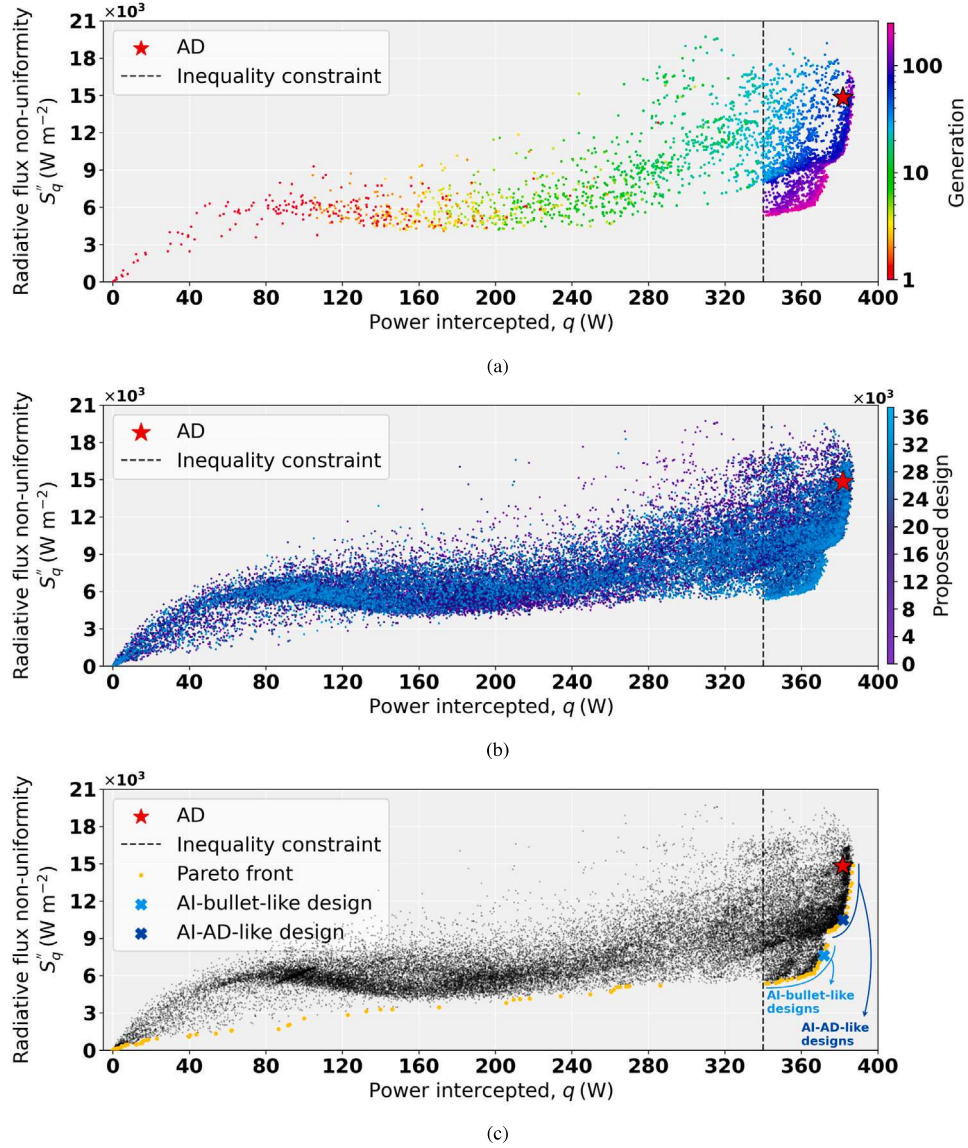


Fig. 12. Radiative flux non-uniformity on the surface of the dual-pipe receiver in tandem with the AI-aided generative-designed secondary receiver vs the power intercepted at the receiver surface: (a) Evolution of the NSGA-II over the generations, (b) representation of the evolution of all designs, and (c) Pareto front of all designs evaluated.

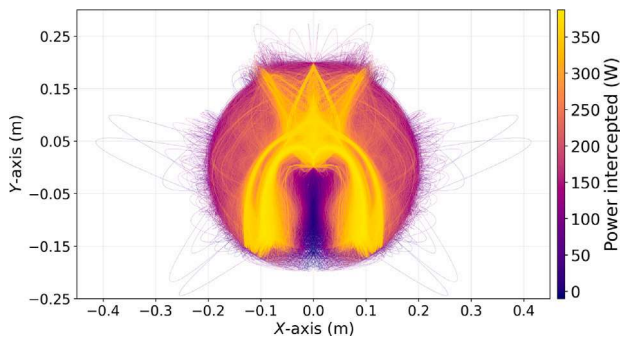


Fig. 13. Superimposition of all secondary reflector geometries generated, with the colour scale referring to the corresponding power intercepted by the dual-pipe receiver.

of the choice of the method employed to parametrise the secondary reflector geometry. This flexible method allows for slight changes in the geometry to reflect the solar radiation to maximise the radiation transfer to the receiver, in contrast to the more constrained analytical

design approach. Since the AI-aided generative design allows to obtain secondary receiver geometries that closely approximate the geometry obtained using the analytical design, the AI-aided methodology is verified and demonstrated as a flexible and automated design approach that overcomes limitations of analytical solutions for non-ideal systems.

The complete set of all generated candidate secondary reflectors is shown in Fig. 13, in which a superimposition of all reflectors is presented.

5.3. Performance comparison

A comparison of the performance of the analytically designed secondary reflector and two selected reflector designs taken from the set of Pareto optimal designs, each representative of a specific design topology. The selected designs from the Pareto front are marked with an 'X' in Fig. 12(c), while their corresponding design geometry is shown in Fig. 14(a).

The AI-AD-like Pareto sub-region shows a high variance with regards to the heat flux distribution uniformity with relatively small change in the power intercepted. Thus, the selected AI-AD-like representative design was chosen based on intercepting a similar amount of

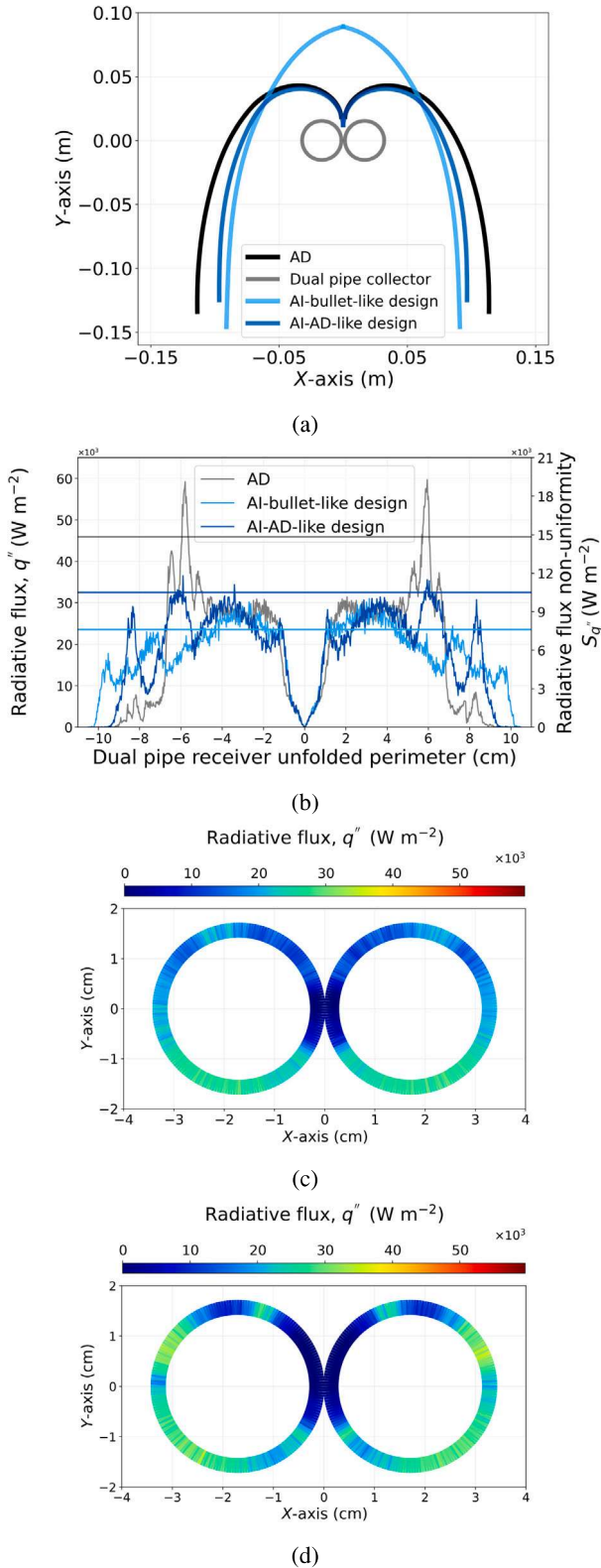


Fig. 14. Comparison of selected characteristics of analytically-designed vs AI-aided generatively-designed secondary reflectors: (a) reflector geometry, (b) unfolded distribution of intercepted radiative flux, (c) intercepted radiative flux distribution at the surface of the dual-pipe receiver with the AI-bullet-like secondary reflector, (d) distribution of intercepted radiative flux at the dual-pipe receiver with the AI-AD-like secondary reflector, (reflectivity 88.3%, sun elevation 90°, azimuth 180°).

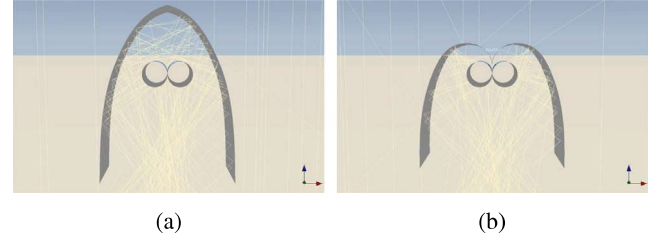


Fig. 15. Rendering of Monte-Carlo raytracing for two selected AI-generated secondary reflectors (sun elevation 90°, azimuth 180°): (a) AI-bullet-like design, and (b) AI-AD-like design.

power as the analytical design and with a better heat flux uniformity. In contrast, the AI-bullet-like Pareto sub-region shows a high variance with regards to the intercepted power with a more contained change in the heat flux distribution uniformity. Thus, the selected AI-bullet-like representative design was chosen based on good compromise between both objective functions, but with more weighting factor on the power intercepted, i.e. to be close to the analytical solution.

The LFR system with the AI-AD-like secondary reflector design achieves the power intercepted at the dual-pipe receiver of $q_{\text{int}} = 381.73 \text{ W}$, with the flux non-uniformity $S_{q''} = 7.6 \text{ W m}^{-2}$. It indicates a smaller flux non-uniformity as compared to the case with the analytically designed reflector, which in turn is characterised by $S_{q''} \approx 15 \text{ W m}^{-2}$. Consequently, the achieved intercepted power and flux non-uniformity for the case with the AI-bullet-like reflector are 371.84 W (corresponding to 97.5% of the power intercepted for the analytical design) and 10.5 W m^{-2} , as shown in Fig. 14(b).

Fig. 15 shows close-up views of both reflectors as simulated in Tonatiuh++ for the reduced length collector and at the same conditions. A small proportion of the casted rays are visualised to show the distribution of rays inside the secondary reflector before being intercepted at the dual-pipe collector surface, in the secondary reflector itself, or reflected back to the environment.

6. Conclusions

A coupled AI and generative design methodology has been employed to optimise the design of secondary reflectors of linear Fresnel collectors. The optimal AI-aided generated receivers are compared against an analytically-derived secondary reflector design, which was used as a figure of merit for the comparison. The proposed methodology allowed to obtain reflector designs that are in very close agreement to the one obtained the analytical solution based on non-imaging optics principles. Furthermore, it also generated reflector designs that outperformed the latter in terms of radiative flux distribution uniformity. Unlike the analytical solution based design, the presented approach can integrate more accurate optical models, such as the sun shape, the atmospheric attenuation, and accurate material properties.

The optimisation presented in this paper required approximately 120 h of computational time on a commercial laptop equipped with i7-1355U CPU (CPU clock 1.7 GHz) and 32 GB of RAM, using 10 CPU cores. Employing different metaheuristic algorithms and hyperparameter settings can be useful to accelerate convergence of the workflow, and thus reduce the computational costs.

Although the methodology employed here is applied to the geometrical design optimisation of secondary reflectors of linear Fresnel collectors, the workflow can be applied to other concentrating solar thermal components, such as the optimisation of cavity and external receivers for solar tower systems.

CRediT authorship contribution statement

Jorge Moreno García-Moreno: Writing – original draft, Visualization, Software, Methodology, Formal analysis, Data curation, Conceptualization. **Alaric Christian Montenon:** Writing – original draft, Visualization, Validation, Methodology, Formal analysis, Data curation, Conceptualization. **Mihalis A. Nicolaou:** Writing – review & editing, Supervision, Funding acquisition, Conceptualization. **Wojciech Lipiński:** Writing – review & editing. **Kypros Milidonis:** Writing – review & editing, Supervision, Project administration, Investigation, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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